
Probe Invariance in Code Models Is Largely Lexical

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Abstract

1 A linear probe on a code model’s residual stream recovers a variable’s *role* almost as
2 well in a language it was not fitted on as in the one it was: across six ordered pairs of
3 Python, JavaScript and PHP, matched problem by problem, transfer moves -0.012
4 macro-F1 at a boundary where a model-free character n -gram classifier loses 0.126.
5 Read at face value that says role representations are language-independent. It is
6 largely an artifact. The probe pools activations over the occurrence’s own tokens, so
7 it reads the variable’s name. The baseline is forbidden to. Pooling only surrounding
8 tokens, with the forward pass unchanged, costs the probe 61% of its margin over
9 an untrained network of the same architecture and pooling ($+0.269 \rightarrow +0.105$)
10 and multiplies its boundary effect fivefold ($-0.012 \rightarrow -0.061$), leaving it no
11 flatter than the untrained network itself (-0.064). The invariance was mostly the
12 identifier. We report this as a caution for cross-lingual probing of code models,
13 together with the model-free comparator such studies have lacked.

14 1 Introduction

15 Probes are how we ask what a code model represents, and cross-lingual probes ask whether what it
16 represents is tied to the language it saw. Prior work has fitted probes on one programming language
17 and evaluated on another [Hernández López et al., 2026], found language-specific components that
18 can be removed from code embeddings [Utpala et al., 2024], and shown that alignment between
19 languages on parallel programs is uneven [Kargaran et al., 2025]. That literature has an untrained-
20 network floor but no model-free one: no non-neural classifier is fitted on the same items, so a probe’s
21 transfer curve has nothing to be read against except itself.

22 We supply that comparator, and it changes the reading. A *variable role* [Sajaniemi, 2002], such as
23 accumulator or index key, is a property of what a variable does rather than of its name or its language,
24 so it is a case where a language-independent representation is plausible on its face. Fitting a role
25 classifier on one language and testing on another, the probe looks strikingly invariant beside a surface
26 baseline that is not.

27 The comparison is unfair in a way that turns out to decide the result. The probe pools activations over
28 the occurrence’s own tokens and therefore reads the identifier, which the surface baseline has masked
29 out. Removing that asymmetry removes most of the effect. What survives is a smaller finding about
30 the surface and a methodological caution about the probe.

31 2 Setup

32 **Corpus.** XLCoST [Zhu et al., 2022] provides parallel solutions keyed by a problem identifier stable
33 across languages. Only Python, JavaScript and PHP share problems at usable rates. No other pair
34 in XLCoST reaches 500 shared problems, including same-family pairs such as C# and Java, which
35 share 175 (Appendix A). We restrict all three languages to the 869 problems all three share, so every

Table 1: Cross-lingual role transfer, macro-F1 averaged over three roles. *Close* is the two directions between JavaScript and PHP. *Python* is the four transfers where Python is one of the two languages. Each probe row is paired with an untrained network at its own pooling.

	close	Python	effect
Surface n -gram, name masked (no model)	0.929	0.804	−0.126
Probe, span-pooled (reads the name)	0.852	0.840	−0.012
untrained, span-pooled [†]	0.590	0.568	−0.022
Probe, context-pooled (no name)	0.630	0.569	−0.061
untrained, context-pooled [†]	0.527	0.464	−0.064

[†] The context-pooled floor was re-run after its original artifacts were lost and awaits commitment; the span-pooled floor is committed but measured on the earlier capped sample rather than the intersection. See Appendix F.

ordered pair scores identical problem ids and a difference between cells cannot be a difference in subject matter. Roles are extracted per occurrence and split by a frozen hash of the problem id. One corpus asymmetry deserves note: XLCoST distributes Python through a formatted mirror and the other two languages through a detokenized one, so the Python boundary is also a formatting boundary. It does not reach the classifier: the character analyser collapses whitespace before forming n -grams, and equalising whitespace on both sides moves transfer by less than 0.001 on every decisive pair.

The four conditions. All are scored on the same occurrences.

- **Surface n -gram.** A character n -gram classifier over the context around an occurrence with every instance of the variable’s name replaced by a placeholder, fitted on the source language and applied to the target. No model. We fix one masked-context variant in advance rather than taking the best per cell. The per-cell maximum is selected on the evaluation fold, and every fixed variant gives a *larger* boundary effect than it.
- **Probe, span-pooled.** Logistic regression on residual activations mean-pooled over the occurrence’s own tokens, layer chosen on source-language validation data. This reads the identifier.
- **Probe, context-pooled.** The same, pooling the 16 tokens either side of the occurrence and *excluding* its own. The forward pass is byte-identical, and only what is read changes. Masking the name in the source would also change the input distribution.
- **Untrained.** The same architecture and tokenizer with randomly initialised weights, run at both poolings. A probe score means little without a bound on what architecture and labels alone supply [Hewitt and Liang, 2019, Zhang and Bowman, 2018, Belinkov, 2022], and a context-pooled probe must be read against a context-pooled floor: the two poolings read different token sets, so their achievable scores differ for reasons unrelated to training.

Models are Qwen2.5-Coder-1.5B [Hui et al., 2024] and, for the untrained control, the architecture of the base model it is continued from [Qwen et al., 2024]. Confidence intervals resample whole problems, never occurrences, since occurrences inside one program share a forward pass.

3 The surface baseline depends on the language

The n -gram baseline reaches 0.929 between JavaScript and PHP without a model and without seeing the variable’s name, and loses 0.126 on transfers to or from Python: 95% CI [−0.160, −0.092], resampling the 427 test problems. The effect is not uniform across roles. Iterator alone gives −0.225, driven by cells that fall to 0.548–0.577 on transfers into Python, while accumulator and index_key together give −0.059. Iterator is also the role whose extractors agree least across the boundary (its labels come from an AST for Python and regular expressions for the other two) and whose intervals are twice as wide, with a minority class of ten instances in some cells. We therefore treat the aggregate as an upper bound on how much of the surface effect is about language rather than labelling, and note that the probe-side conclusions do not depend on it: the identifier collapse holds for every role.

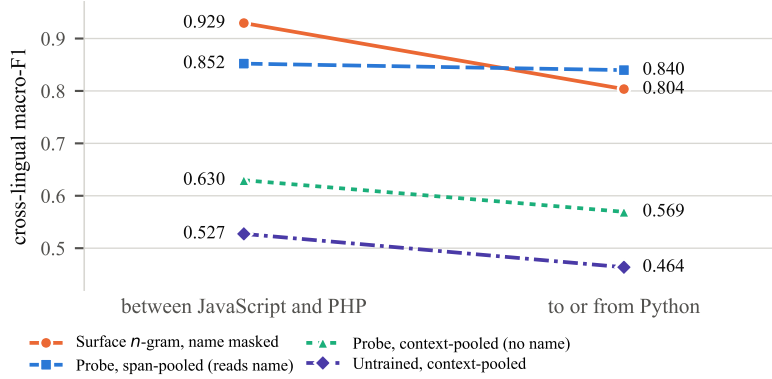


Figure 1: Table 1 as a picture. The surface baseline tracks the boundary; the span-pooled probe does not, but it reads the variable’s name. Excluding the name drops the probe to just above its untrained floor, and both slope downward together.

4 The probe looks invariant, and mostly should not

Read against that, the span-pooled probe is remarkable: it moves -0.012 , an order of magnitude less, and clears an untrained network of the same architecture by $+0.269$. That is the result the obvious reading calls language-independent role representation.

It does not survive removing the identifier. The probe pools over the occurrence’s own tokens, so the variable’s name is inside what it reads, while the baseline it is being compared against has that name masked. Pooling only the surrounding tokens, with the forward pass unchanged, the probe falls from 0.844 to 0.589 overall (Figure 1). Against its *matched* untrained floor of 0.485 that is a margin of $+0.105$, against $+0.269$ before: 61% of what the probe knew came from the identifier.

So did the invariance. The boundary effect grows from -0.012 to -0.061 , and the untrained network at the same pooling moves -0.064 . With the identifier removed the trained probe is no flatter than an untrained one.

The same shift shows up within languages. In-domain, the span-pooled probe scores 0.898, and excluding the name costs it 0.127 (0.771), against an untrained in-domain of 0.623 at the same pooling. So the name is not only the transferable part of the signal, it is a substantial part of the signal everywhere, and the cross-lingual result is the within-language dependence made visible by a boundary the name happens to cross freely. The apparent language-independence was not a property of the role representation. It was a property of variable names, which parallel implementations of the same problem tend to share.

What remains. A margin of $+0.105$ over a matched floor is small but consistent in both groups, so the model does encode something about roles beyond the identifier. We do not claim it is language-independent: at that pooling its boundary effect is indistinguishable from an untrained network’s. The advantage is also paired, not only average: over the same occurrences the surface baseline beats the context-pooled probe in 14 of 18 cells with a problem-clustered interval excluding zero (Appendix F).

Related work. Cross-lingual analysis of code models has established language-specific components that can be removed from embeddings [Utpala et al., 2024], cross-language probe transfer with untrained-network floors [Hernández López et al., 2026], and uneven alignment between these very languages on parallel programs [Kargaran et al., 2025]. None of it fits a model-free classifier on the same items, which is the condition that exposed the asymmetry here. On the probing side we inherit the control-task and probe-validity tradition [Hewitt and Liang, 2019, Voita and Titov, 2020, Belinkov, 2022]. Our contribution to it is the observation that these controls hold the probe’s *input* fixed, and so cannot detect a label correlate carried by the input itself.

5 Discussion

The finding is methodological, and it generalises past this corpus. Any probe that pools over a span containing an identifier can read that identifier, and in parallel corpora identifiers are among the most language-invariant things present. A cross-lingual probing result obtained that way will overstate representational invariance by however much the naming happens to agree, and no untrained-network control detects this: the untrained network reads the same identifier and is defeated by the same confound. Only a comparator that cannot see the name, or a pooling that excludes it, separates the two.

That is also why the model-free arm matters. The surface baseline is the thing that made the asymmetry visible, because it was the only condition with the identifier withheld. Prior cross-lingual probing of code models [Hernández López et al., 2026, Utpala et al., 2024, Kargaran et al., 2025] has untrained-network floors but no model-free comparator, and would not surface this. The concern generalises the standard caution that a probe reports what is decodable rather than what the model uses [Belinkov, 2022, Voita and Titov, 2020]: here part of what is decodable is the input itself, and it survives the control usually relied on.

Reconciling with prior alignment results. Kargaran et al. [2025] find Python the *least* aligned of these three languages on parallel programs, which sits oddly beside a span-pooled probe that barely moves across the Python boundary. The two measure different things, whole-snippet retrieval against role decodability, but the tension dissolves entirely under our control: once the identifier is excluded, our probe shows a boundary effect after all, in the direction their alignment scores predict. What looked like a disagreement between measurements was the confound, and removing it brings the two studies into agreement, which is itself modest evidence that both are measuring something real about the Python boundary rather than artifacts of their respective designs.

What would change our reading. Three observations would. A context-pooled probe at larger scale clearing its floor by a margin that grows rather than shrinks would show role abstraction emerging with capacity. A pooling that excludes the identifier but retains its dependency structure (e.g. masking the name at the input while pooling the span) recovering the flat curve would localise the confound to the read-out rather than the representation. And a corpus whose parallel implementations do *not* share naming conventions showing a flat span-pooled probe would break our explanation outright. Each is a concrete follow-up, and the second is one GPU session on the released pipeline.

For language-as-design-axis studies. The practical upshot for this workshop’s programme is a measurement rule rather than a finding about code: when the medium is manipulated, audit what the measuring instrument reads before crediting the representation. Parallel corpora, the natural substrate for medium manipulations, are precisely where instrument-visible invariants such as names, entities and numbers concentrate. A probe that can see them will report the manipulation harmless; an instrument denied them, model-free or otherwise, is the cheap check that says whether the harmlessness is real.

Limitations. Three languages, of which one is the boundary, so the contrast has two points and no gradient, and Appendix A shows XLCOST admits no wider version: no pair outside this triangle shares even 500 problems, including same-family pairs such as C# and Java. One model at 1.5B, so nothing here speaks to whether the name-free margin grows with scale. The identifier-lift figures are aggregated over three roles and we do not report them per role. Iterator’s labels come from a different extractor path than Python’s, its minority class falls to ten instances in some cells, and it carries most of the surface effect. The accumulator and index_key figures are the more trustworthy ones. Context pooling excludes the identifier but not everything correlated with it (nearby uses of the same variable, or names of neighbouring variables that co-vary with it), so +0.105 is an upper bound on identifier-free role information. The context window of 16 tokens is a choice we did not sweep. Finally, this is a probing result: it says what is decodable, not what the model uses, and no causal claim follows from it.

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Table 2: Shared problem identifiers between every pair of XLCOST languages, train split. Matched cross-lingual transfer needs a pair to share enough problems to hold out a test fold; only the three bold cells clear 500. No pair outside Python/JavaScript/PHP does, including same-family pairs.

	C++	C#	Java	JavaScript	PHP	Python
C++	—	115	122	75	17	74
C#	115	—	175	75	14	62
Java	122	175	—	85	11	66
JavaScript	75	75	85	—	1,529	2,953
PHP	17	14	11	1,529	—	1,145
Python	74	62	66	2,953	1,145	—

Table 3: All eighteen transfer cells for Qwen2.5-Coder-1.5B. *Surface* is the strongest masked-context n -gram feature; *name* uses the variable’s name alone; *probe* is the residual-stream probe. Majority and shuffled-label controls sit near chance throughout, and ρ is the macro-F1 movement from one test occurrence changing its prediction.

role	pair	n	surface	name	probe	major.	shuf.	ρ
accumulator	JavaScript→PHP	1,976	0.951	0.897	0.904	0.361	0.470	0.0005
accumulator	JavaScript→Python	3,341	0.813	0.873	0.875	0.415	0.473	0.0004
accumulator	PHP→JavaScript	1,903	0.954	0.897	0.918	0.372	0.490	0.0005
accumulator	PHP→Python	1,377	0.782	0.824	0.833	0.276	0.439	0.0008
accumulator	Python→JavaScript	3,294	0.785	0.874	0.918	0.370	0.504	0.0003
accumulator	Python→PHP	1,478	0.736	0.845	0.874	0.271	0.466	0.0007
index_key	JavaScript→PHP	1,976	0.940	0.834	0.897	0.384	0.478	0.0005
index_key	JavaScript→Python	3,341	0.859	0.787	0.883	0.330	0.497	0.0003
index_key	PHP→JavaScript	1,903	0.964	0.838	0.880	0.376	0.516	0.0005
index_key	PHP→Python	1,377	0.839	0.782	0.854	0.387	0.515	0.0008
index_key	Python→JavaScript	3,294	0.871	0.822	0.915	0.314	0.470	0.0003
index_key	Python→PHP	1,478	0.858	0.838	0.874	0.419	0.459	0.0008
iterator	JavaScript→PHP	1,976	0.937	0.862	0.906	0.448	0.488	0.0008
iterator	JavaScript→Python	3,341	0.774	0.733	0.939	0.444	0.472	0.0005
iterator	PHP→JavaScript	1,903	0.965	0.873	0.851	0.446	0.478	0.0008
iterator	PHP→Python	1,377	0.576	0.788	0.865	0.428	0.462	0.0010
iterator	Python→JavaScript	3,294	0.790	0.769	0.918	0.466	0.484	0.0007
iterator	Python→PHP	1,478	0.741	0.788	0.902	0.475	0.438	0.0020

198 A Pairwise problem overlap in XLCOST

199 B Full transfer table

200 C Transfer mechanism

201 D Language geometry

202 **The typing axis is not privileged.** Probing the same activations under every alternative 4-versus-3
203 partition of the seven languages ($\binom{7}{4} = 35$ groupings minus the real one, so 34 controls) gives
204 best-layer macro-F1 0.9959 on average (min 0.9918, max 1.0000), against 1.0000 for the true
205 static/dynamic split; 5 of the 34 controls match or exceed it. Any grouping of these languages is
206 near-perfectly decodable, so the decodability of the typing split is evidence that language identity is
207 encoded, not that type discipline is. At layer 4, where $|r|$ peaks at 0.941, the remaining components
208 carry almost none of the label: PC2 $r = -0.014$, PC3 $r = +0.026$, PC4 $r = +0.171$.

209 E Causal interventions

210 **Scope.** Causal interventions were run for the *boolean* role over 5 languages (C++, Java, JavaScript,
211 PHP, Python), 3 models, and 3 intervention modes (ablate, patch, steer), giving 375 layer-wise

Table 4: Probe transfer per model, averaged within each group. The untrained network shares Qwen2.5-1.5B’s architecture and tokenizer with randomly initialised weights; it is the floor a representational claim must clear, and it is itself flat.

model	close pair	Python pairs	all	vs. untrained
Qwen2.5-1.5B	0.899	0.888	0.892	+0.316
Qwen2.5-Coder-1.5B	0.893	0.887	0.889	+0.314
Qwen2.5-1.5B (untrained)	0.590	0.568	0.575	—

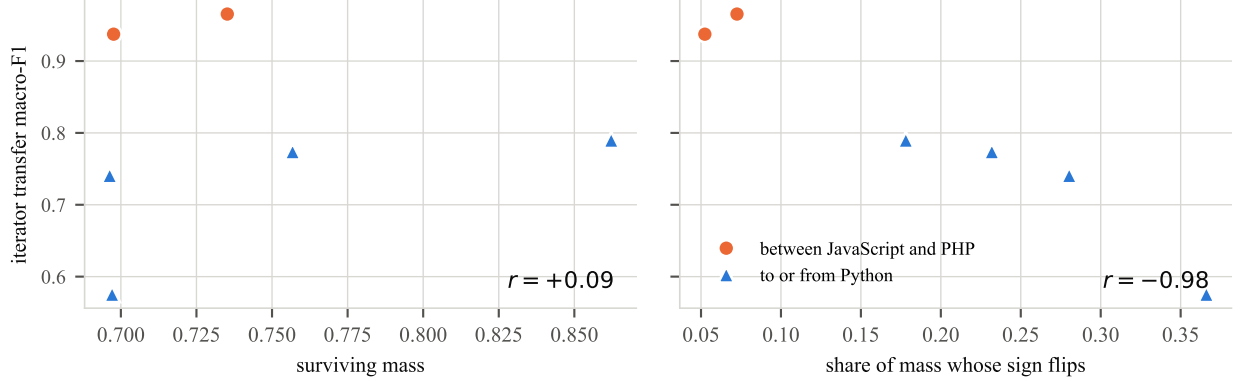


Figure 2: Two candidate explanations for the transfer gap, over the six ordered pairs. Left: how much of the source classifier’s discriminative mass the target realises, which is essentially unrelated to transfer. Right: the share of that mass whose sign reverses between a source-fitted and a target-fitted classifier, which tracks it closely. The cues are present in both groups; what changes is where they point.

212 measurements. They are reported here rather than in the main text because they were run *within*
213 languages and on a role outside the three this paper transfers, so they cannot support its central claim.
214 Specificity is $|\text{clean} - \text{intervened}|$ divided by the same quantity for a random-position control: how
215 much of the effect is specific to the variable’s own token positions rather than to editing anything at
216 all. Two columns are given because the layer with the largest effect is generally not the layer with the
217 best specificity, and quoting either alone misleads.

218 F Reproducibility

219 Every figure is regenerated from committed result files, and a regression test recomputes each of
220 them from those files rather than quoting them, with one exception. The context-pooled untrained
221 row of Table 1 comes from a completed re-run whose per-cell artifacts are pending commitment to
222 `results/lp4fm/masked_probe/`; the span-pooled untrained row is committed, but was measured
223 on the earlier capped sample rather than the three-way intersection the other rows use, so the +0.269
224 margin pairs conditions from two samples and should be read as approximate. Activation extraction is
225 the only step needing an accelerator. The probe is logistic regression on activations mean-pooled over
226 either the occurrence’s span or the 16 tokens either side of it excluding the span; the surface baseline
227 is a `char_wb` tf-idf vectoriser with $n \in [2, 4]$ followed by logistic regression, fitted on the source
228 language only. Pooling and initialisation are recorded in each store’s identity, so a context-pooled
229 store cannot be confused with a span-pooled one downstream.

Table 5: Transfer mechanism for the iterator role. *Surviving* is the share of the source classifier’s discriminative mass realised in the target; *flip* is the share of that mass whose sign reverses. Surviving mass is uncorrelated with transfer; flipped mass predicts it almost exactly. Source-weighted variants are given for robustness.

pair	F1	surviving	flip	agree	flip (src-wt.)
PHP→JavaScript	0.965	0.735	0.072	+0.895	0.072
JavaScript→PHP	0.937	0.698	0.052	+0.945	0.068
Python→JavaScript	0.790	0.862	0.178	+0.695	0.211
JavaScript→Python	0.774	0.757	0.232	+0.811	0.241
Python→PHP	0.741	0.696	0.280	+0.528	0.367
PHP→Python	0.576	0.697	0.366	+0.476	0.346

Table 6: Masking an entire syntactic character class on both sides of the transfer. No class accounts for more than 0.021 of the 0.39 gap, so the realignment is distributed rather than carried by block delimiters or statement terminators.

pair	masked class	macro-F1	Δ
PHP→JavaScript	terminator	0.9496	-0.0159
PHP→JavaScript	brace	0.9645	-0.0010
PHP→JavaScript	bracket	0.9711	+0.0057
PHP→JavaScript	paren	0.9609	-0.0045
PHP→JavaScript	operator	0.9441	-0.0213
PHP→Python	terminator	0.5786	+0.0030
PHP→Python	brace	0.5750	-0.0006
PHP→Python	bracket	0.5733	-0.0023
PHP→Python	paren	0.5947	+0.0191
PHP→Python	operator	0.5801	+0.0045

Table 7: Point-biserial correlation between PC1 of last-token residual activations and a static/dynamic typing label, by layer, with PC1’s explained-variance ratio. The sign of r is arbitrary (PCA fixes the axis only up to reflection, and each layer is refit independently); magnitude is what carries meaning.

layer	$ r $	EVR	layer	$ r $	EVR
0	0.883	0.481	15	0.894	0.309
1	0.898	0.447	16	0.882	0.303
2	0.914	0.391	17	0.874	0.296
3	0.939	0.373	18	0.873	0.289
4	0.941	0.370	19	0.873	0.299
5	0.862	0.369	20	0.903	0.291
6	0.828	0.361	21	0.886	0.276
7	0.844	0.342	22	0.859	0.265
8	0.896	0.339	23	0.842	0.252
9	0.919	0.333	24	0.812	0.242
10	0.880	0.327	25	0.774	0.253
11	0.893	0.302	26	0.694	0.255
12	0.917	0.292	27	0.732	0.237
13	0.904	0.314	28	0.576	0.242
14	0.898	0.309			

Table 8: Causal interventions on boolean-flag occurrences. *Peak-effect spec.* is specificity at the largest-effect layer; *best spec.* is the maximum over layers, with that layer named.

language	mode	model	peak-effect spec.	best spec. (layer)
C++	ablate	qwen2515b	4.5×	31.9× (L4)
C++	ablate	qwen25coder15b	4.4×	76.2× (L20)
C++	ablate	starcoder27b	4.3×	36.5× (L4)
C++	patch	qwen2515b	3.6×	7.9× (L20)
C++	patch	qwen25coder15b	3.6×	6.9× (L8)
C++	patch	starcoder27b	8.1×	10.4× (L16)
C++	steer	qwen2515b	45.0×	169.8× (L4)
C++	steer	qwen25coder15b	598.5×	598.5× (L24)
C++	steer	starcoder27b	6.9×	341.1× (L4)
Java	ablate	qwen2515b	28.5×	86.0× (L12)
Java	ablate	qwen25coder15b	12.3×	62.4× (L12)
Java	ablate	starcoder27b	27.0×	27.0× (L4)
Java	patch	qwen2515b	6.1×	7.7× (L12)
Java	patch	qwen25coder15b	5.9×	7.7× (L12)
Java	patch	starcoder27b	7.5×	8.6× (L12)
Java	steer	qwen2515b	72.1×	849.0× (L8)
Java	steer	qwen25coder15b	71.9×	186.6× (L0)
Java	steer	starcoder27b	252.0×	252.0× (L24)
JavaScript	ablate	qwen2515b	26.7×	26.7× (L16)
JavaScript	ablate	qwen25coder15b	28.1×	116.3× (L24)
JavaScript	ablate	starcoder27b	27.3×	27.3× (L4)
JavaScript	patch	qwen2515b	6.9×	6.9× (L8)
JavaScript	patch	qwen25coder15b	6.6×	7.4× (L12)
JavaScript	patch	starcoder27b	8.1×	8.2× (L16)
JavaScript	steer	qwen2515b	24.5×	76.4× (L8)
JavaScript	steer	qwen25coder15b	65.2×	149.5× (L4)
JavaScript	steer	starcoder27b	11.2×	60.8× (L28)
PHP	ablate	qwen2515b	15.0×	102.4× (L20)
PHP	ablate	qwen25coder15b	26.5×	79.2× (L20)
PHP	ablate	starcoder27b	19.4×	19.7× (L8)
PHP	patch	qwen2515b	7.2×	7.2× (L4)
PHP	patch	qwen25coder15b	4.7×	7.4× (L0)
PHP	patch	starcoder27b	6.3×	9.3× (L16)
PHP	steer	qwen2515b	190.3×	190.3× (L24)
PHP	steer	qwen25coder15b	11.5×	134.0× (L8)
PHP	steer	starcoder27b	8.7×	100.1× (L4)
Python	ablate	qwen2515b	5.8×	36.7× (L24)
Python	ablate	qwen25coder15b	6.8×	33.1× (L16)
Python	ablate	starcoder27b	26.5×	26.5× (L4)
Python	patch	qwen2515b	6.0×	7.2× (L16)
Python	patch	qwen25coder15b	6.7×	7.8× (L16)
Python	patch	starcoder27b	7.9×	9.7× (L12)
Python	steer	qwen2515b	209.1×	827.9× (L16)
Python	steer	qwen25coder15b	1025.8×	1025.8× (L24)
Python	steer	starcoder27b	8.6×	55.5× (L4)

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